

Student Performance Analysis Dashboard

Overarching Insight:

This analysis reveals that **internet access** and **school type** are the most significant drivers of student performance, with notable score gaps. Surprisingly, **study hours** and **attendance**, often considered crucial, show no measurable correlation with overall scores in this dataset. Subject scores remain strong predictors, highlighting the need for targeted academic support rather than solely focusing on generic study habits.

1. Data Loading and Preparation

Cleaned dataset loaded successfully.

Data snippet after loading and preparing additional columns:

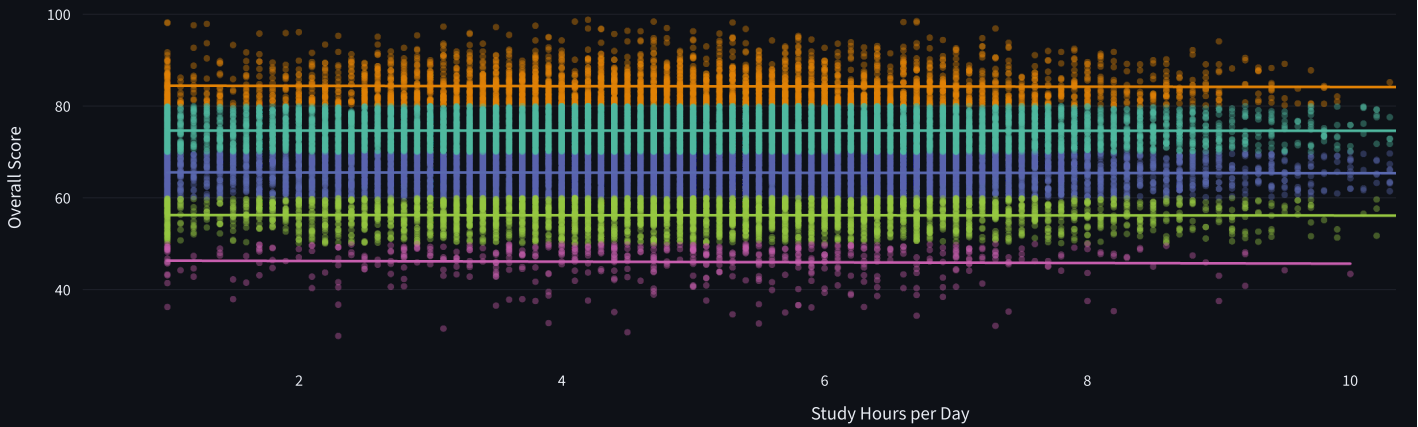
	student_id	age	gender	school_type	parent_education	study_hours	attendance_percentage	internet_access	travel_time	extra_activities	study_method
0	1	20	Female	Private	High School	1.1	93.8	yes	15-30 min	yes	Group Study
1	2	17	Male	Private	Diploma	7.4	91.7	no	30-60 min	yes	Coaching
2	3	19	Male	Private	High School	7	100	no	30-60 min	no	Group Study
3	4	19	Male	Private	Diploma	1.6	79.9	no	15-30 min	no	Self Study
4	5	18	Male	Private	Graduate	1.9	99	yes	15-30 min	no	Self Study

2. Key Performance Insights (Interactive Visualizations)

Q1: Do study hours positively impact overall performance?

The number of hours a student studies per day does **not** have a meaningful positive (or negative) impact on overall score in this dataset, showing an essentially flat trendline (correlation ≈ -0.01).

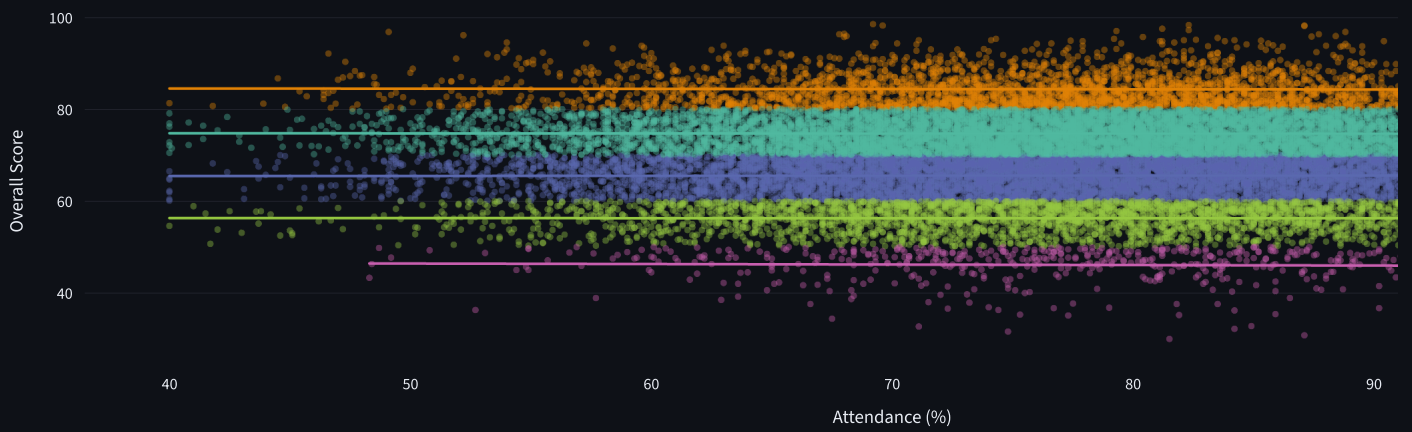
Study Hours vs Overall Score (with trendline)



Q2: Does attendance have a strong positive relationship with scores?

Attendance alone does not strongly drive performance in this dataset, with a correlation of approximately -0.007 with overall scores.

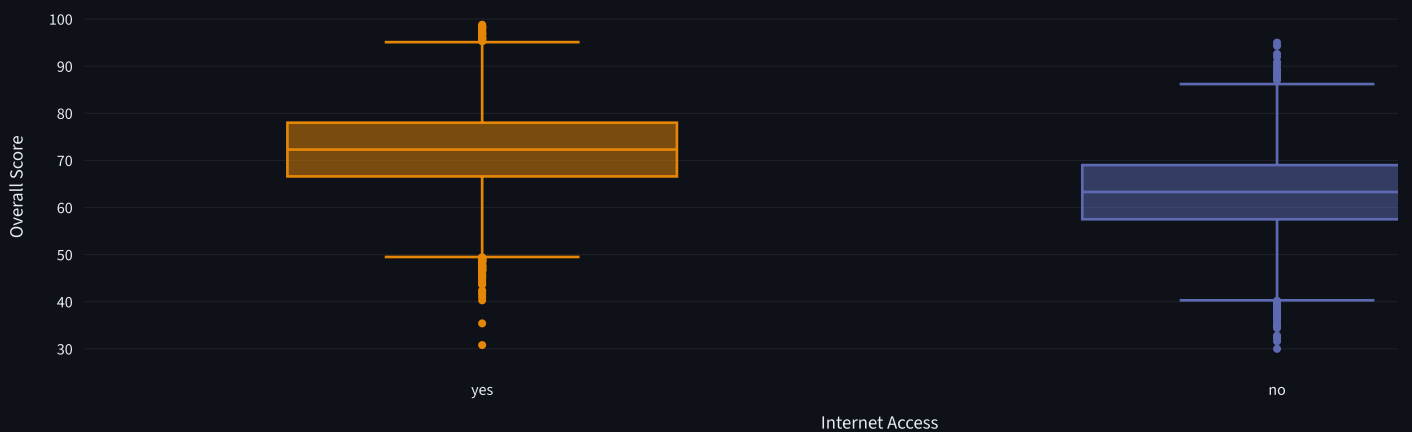
Attendance % vs Overall Score (with trendline)



Q3: Does internet access improve academic outcomes?

Students **with** internet access average ~72.2 overall versus ~63.2 for those **without** — a significant ~9-point gap, making internet access the largest single-factor differentiator in this dataset.

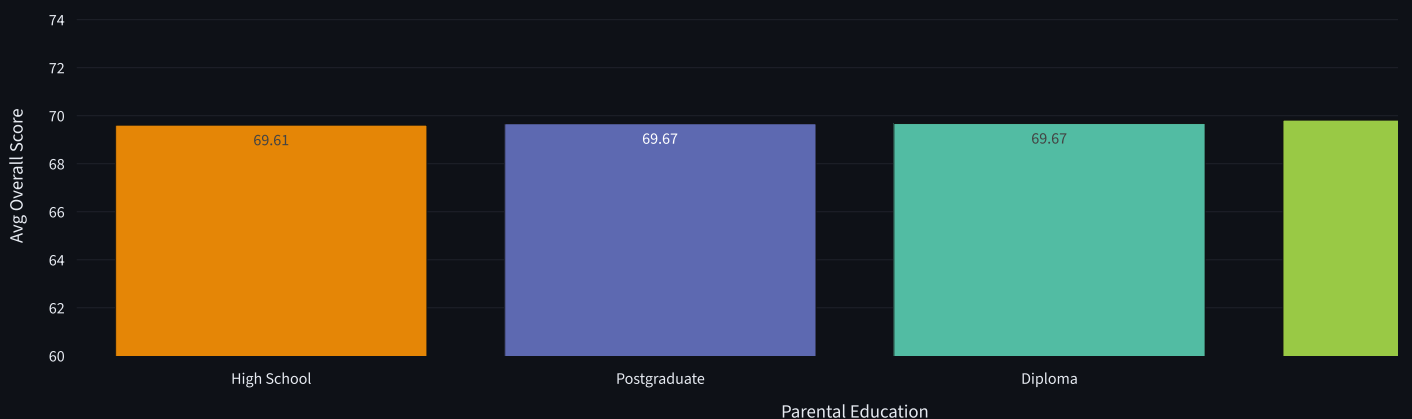
Overall Score by Internet Access



Q4: Does parental education influence student achievement?

Parental education has little practical effect on student achievement, with average overall scores differing by less than half a point across education levels.

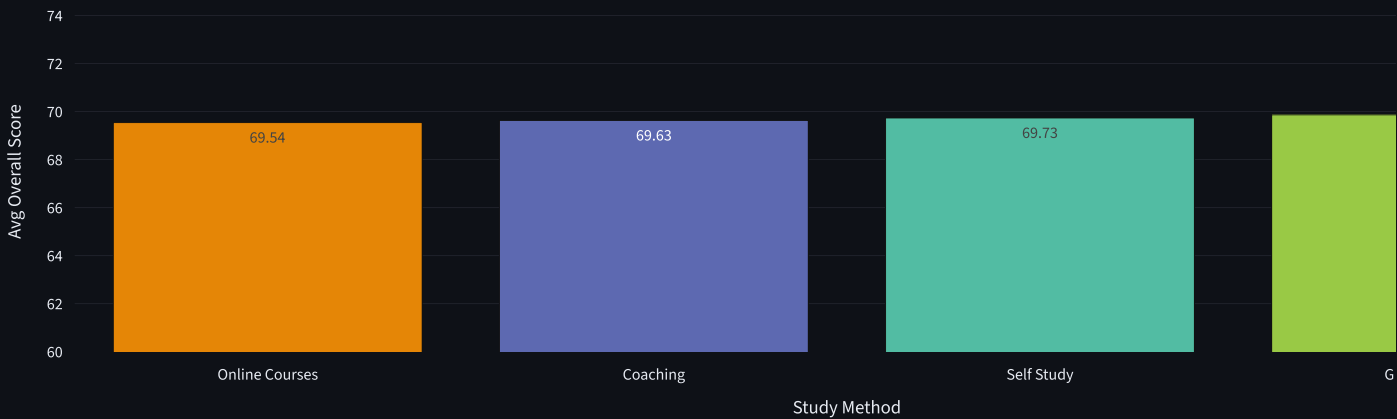
Average Overall Score by Parental Education



Q5: Do certain study methods lead to better scores?

While differences are small, 'Group Study' slightly outperforms other methods, but the overall spread of less than 0.4 points suggests a minor effect of study method on overall scores.

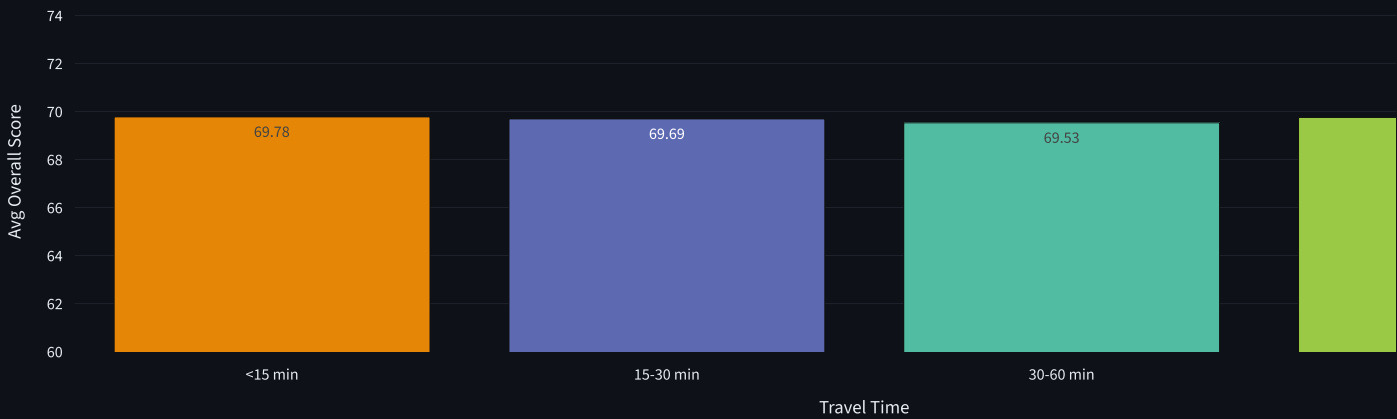
Average Overall Score by Study Method



Q6: Do students with shorter travel times perform better?

Travel time has a negligible impact on overall scores, with differences across categories being less than 0.3 points and no clear performance trend.

Average Overall Score by Travel Time

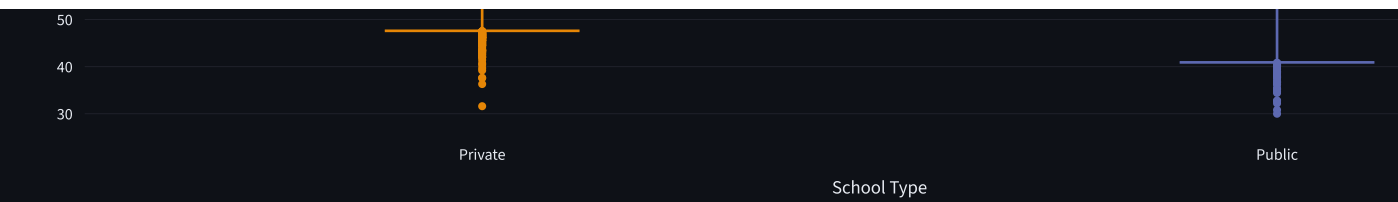


Q7: Does school type impact student performance?

Private school students achieve substantially higher overall scores (~71.6) compared to Public school students (~65.7), indicating a strong 6-point advantage for private schooling.

Overall Score by School Type

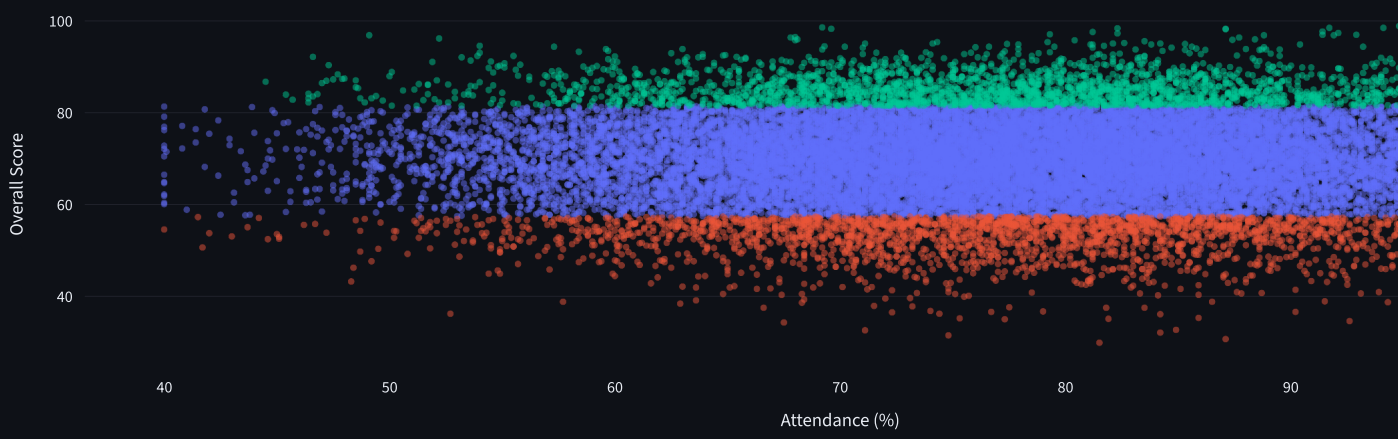




Q8 & Q14: How can top performers be identified & profiled?

Students in the top 10% of overall score (≥ 81.5) are identifiable, allowing for targeted scholarship and recognition programs. Below, we also explore the performance tier breakdown by study method.

Top Performers Highlighted (Overall Score ≥ 81.5)



Q14: Performance Tier Breakdown by Study Method

No single study method guarantees top performance, but 'Group Study' shows a slightly higher proportion of 'Top Performers' compared to other methods.

Performance Tier Breakdown (%) by Study Method



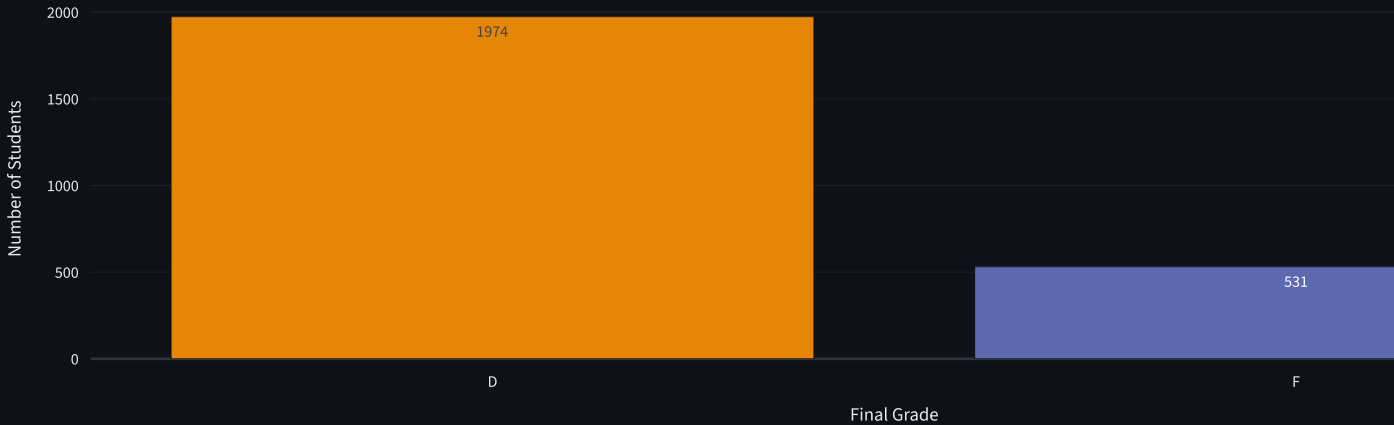
Top 10 candidates for scholarships

	student_id	overall_score	math_score	science_score	english_score	attendance_percentage	study_hours	final_grade
1135	1136	98.8000	100.0000	100.0000	96.5000	94.8000	4.2000	A
22515	1261	98.5000	96.7000	100.0000	98.8000	69.2000	6.7000	A
3824	3825	98.4000	99.4000	96.3000	99.5000	91.7000	4.1000	A
16285	6317	98.4000	100.0000	100.0000	95.3000	94.1000	4.7000	A
12267	12268	98.3000	95.0000	100.0000	100.0000	82.3000	6.6000	A
8785	8786	98.2000	97.0000	97.6000	100.0000	100.0000	1.0000	A
13288	13289	98.2000	100.0000	97.2000	97.3000	69.6000	6.7000	A
23687	10491	98.2000	100.0000	100.0000	94.5000	87.1000	5.3000	A
2603	2604	98.1000	100.0000	100.0000	94.2000	87.1000	1.0000	A
17977	3875	97.9000	96.8000	100.0000	97.0000	96.6000	1.3000	A

Q9: How can low-performing students be targeted for intervention?

Students in the bottom 10% of overall score (≤ 57.4) are identified for targeted academic interventions to improve their performance.

Students Flagged for Intervention, by Final Grade

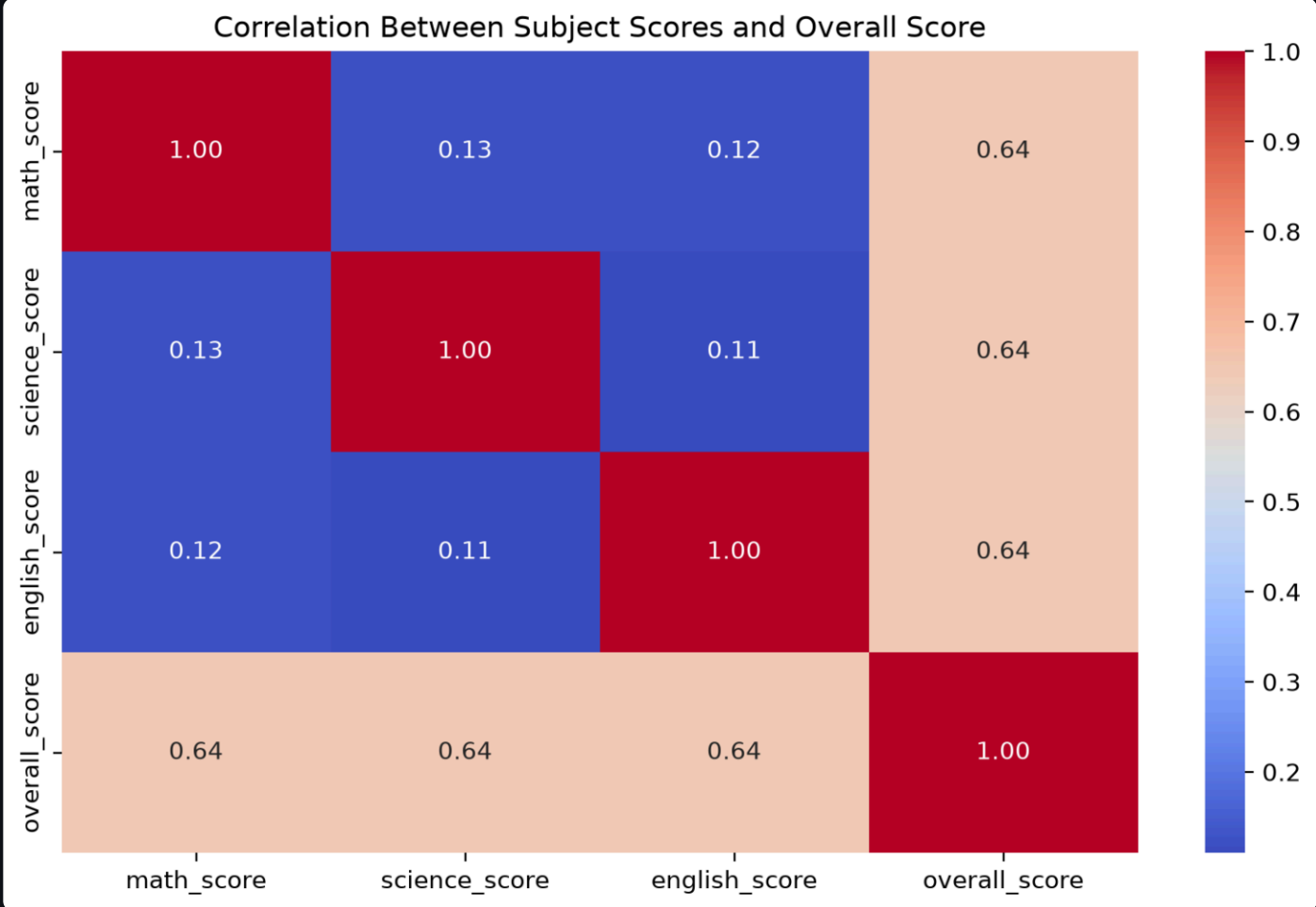


10 lowest scorers needing intervention

	student_id	overall_score	math_score	science_score	english_score	attendance_percentage	study_hours	final_grade
18547	3293	30.0000	42.2000	25.2000	22.6000	81.5000	2.3000	F
10305	10306	30.8000	35.7000	32.3000	24.3000	87.1000	4.5000	F
7763	7764	31.6000	33.9000	20.0000	40.8000	74.8000	3.1000	F
3605	3606	32.2000	22.3000	34.0000	40.3000	84.2000	7.3000	F
12530	12531	32.7000	23.0000	40.6000	34.4000	71.1000	5.5000	F
19713	12738	32.8000	37.3000	38.3000	22.8000	84.9000	3.9000	F
14359	14360	34.4000	31.0000	20.4000	51.8000	67.5000	6.7000	F
11320	11321	34.7000	48.4000	35.7000	20.0000	92.6000	5.3000	F
11012	11013	35.1000	50.2000	32.5000	22.7000	77.3000	5.7000	F
17019	3565	35.2000	39.8000	25.8000	39.9000	81.9000	4.4000	F

Q10: Do Math, Science, and English scores strongly affect the overall score?

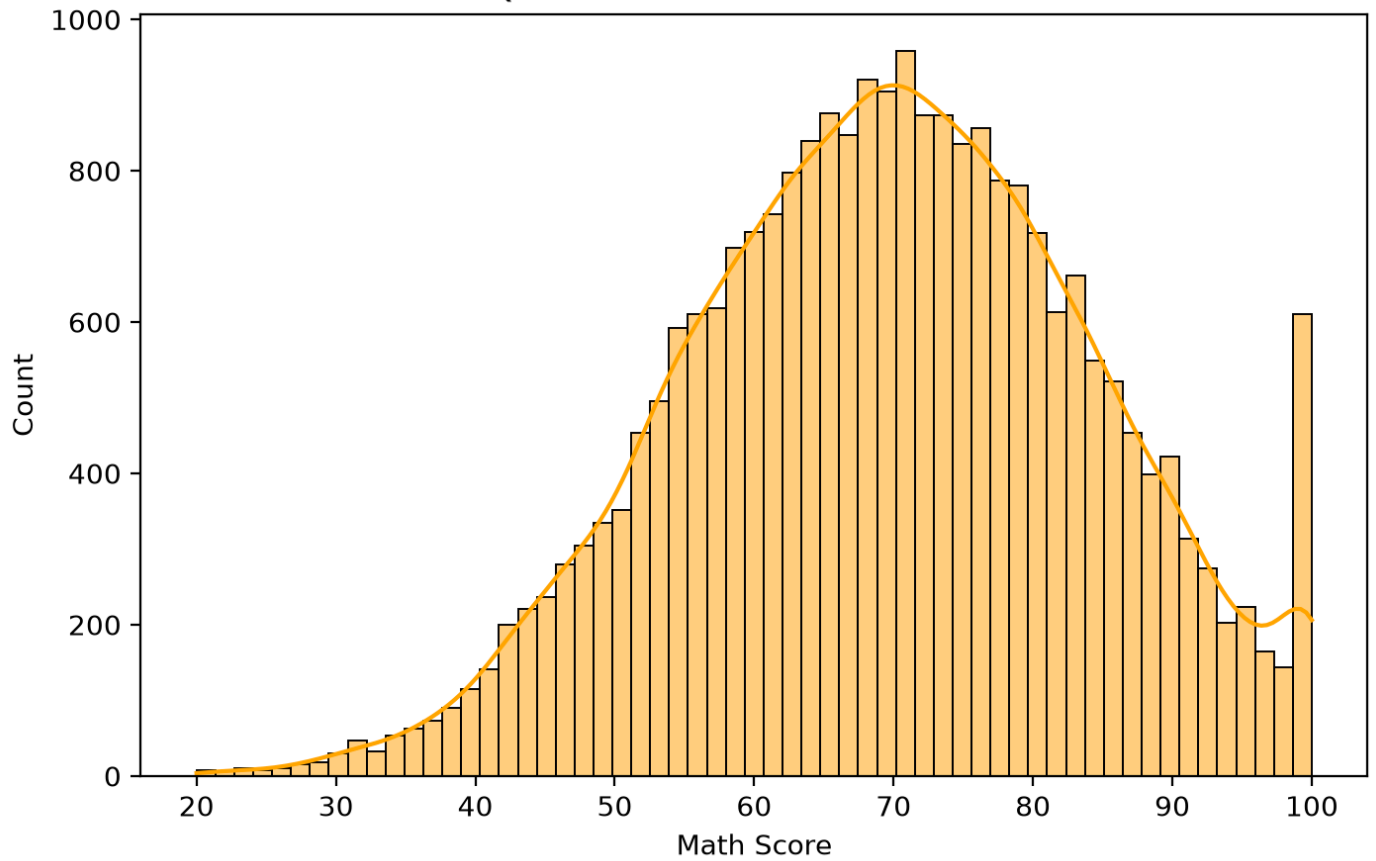
Math, Science, and English scores are very strong predictors, each correlating ~ 0.64 with the overall score, which is essentially their average.



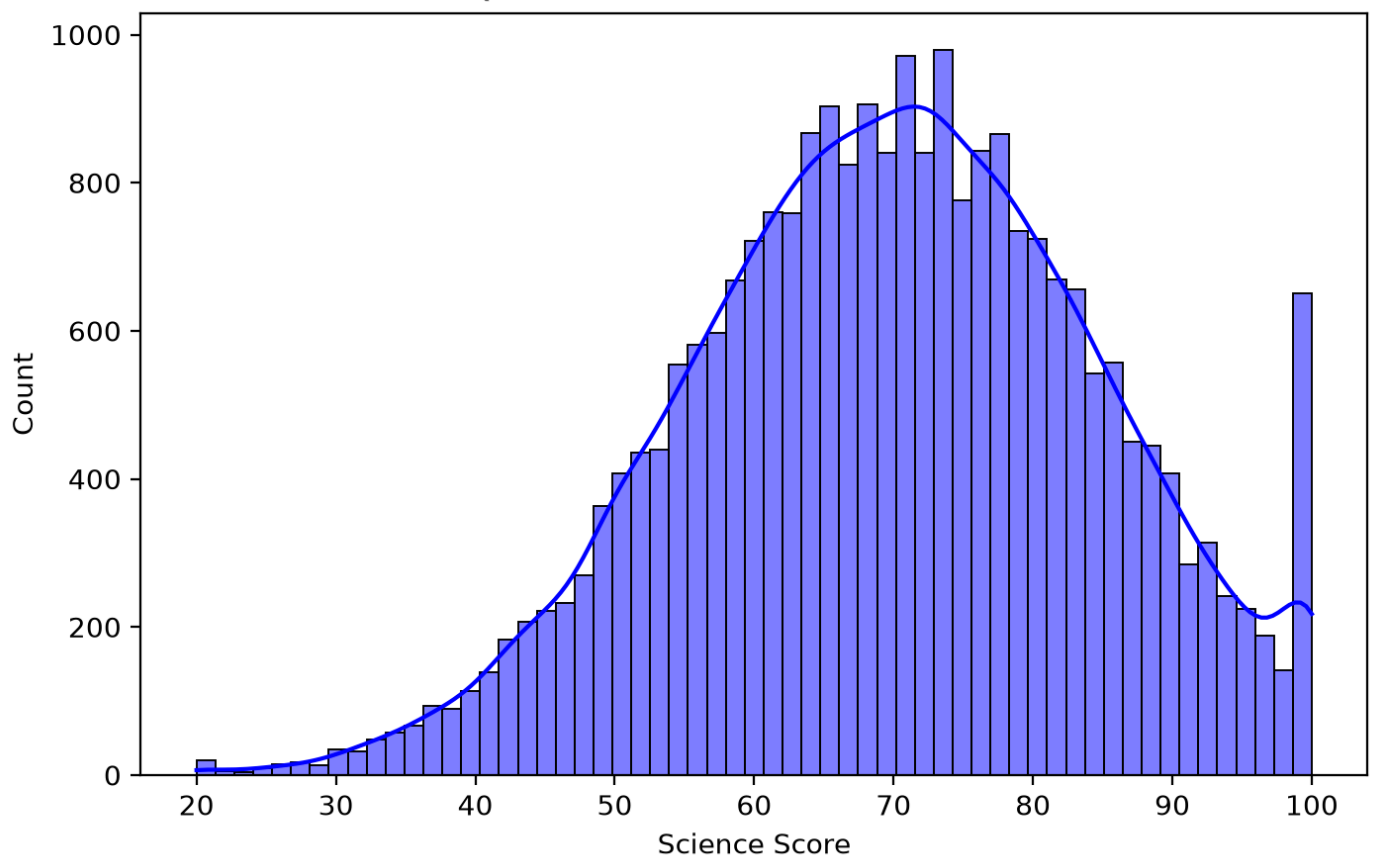
Q11, Q12 & Q13: Distributions of Math, Science, and English scores

- **Q11 (Math):** Math scores are roughly bell-shaped and concentrated in the 60s-70s, indicating most students perform in the average range.
- **Q12 (Science):** Science scores show a similar bell shape, centered slightly above 70, suggesting moderate to strong performance overall.
- **Q13 (English):** English scores are also roughly symmetric around the 60s-70s, indicating a fairly even spread around the average.

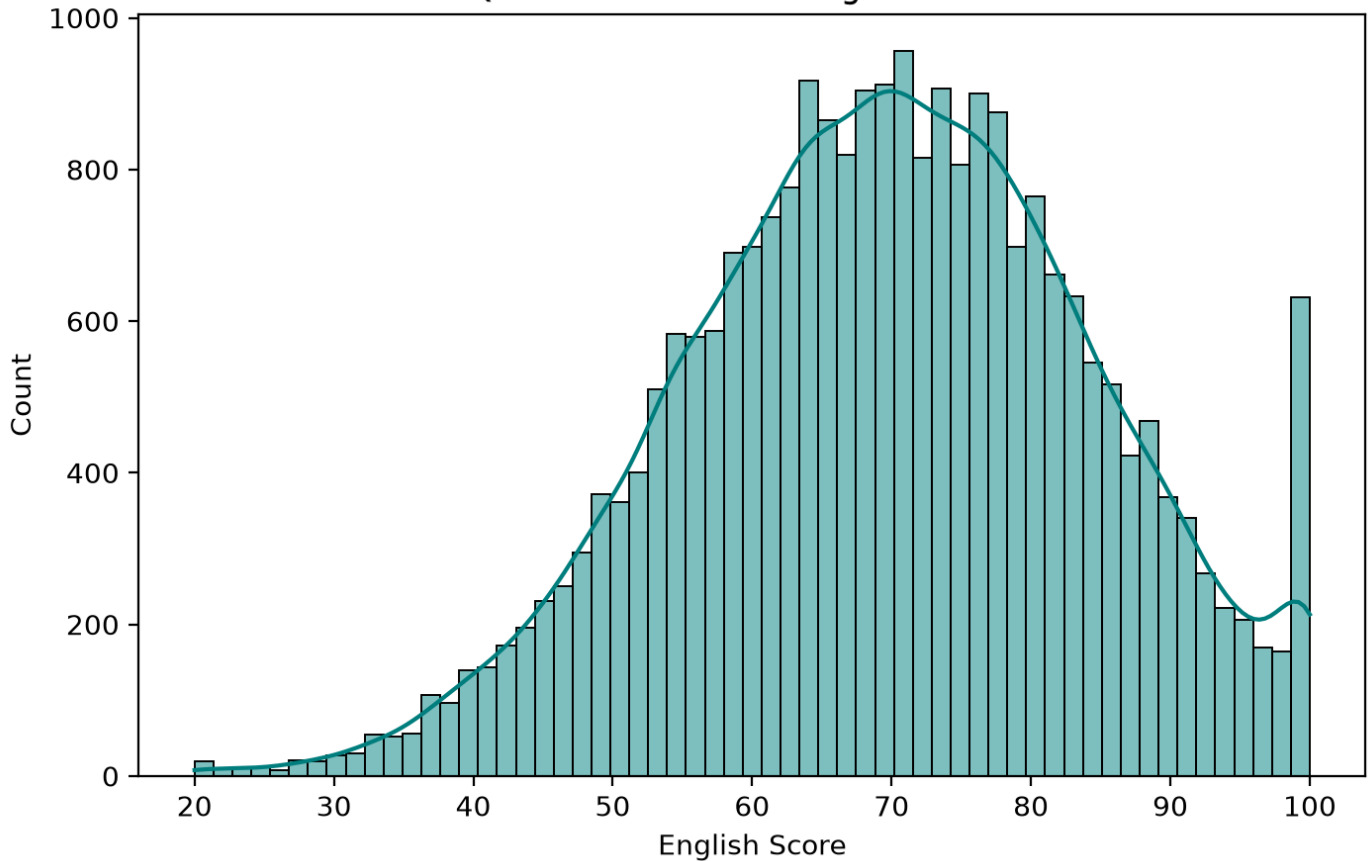
Q11: Distribution of Math Scores



Q12: Distribution of Science Scores



Q13: Distribution of English Scores



Q15: Are subject scores, attendance, and study hours the strongest predictors of overall performance?

Subject scores are the strongest predictors, with attendance and study hours showing virtually no correlation with overall performance in this dataset.

Correlation of Each Feature with Overall Score



Deployment Instructions

To deploy this Streamlit dashboard:

- Save the code:** Save the Python code above as a `.py` file (e.g., `dashboard_app.py`) in the same directory as your `Student_Performance_Analysis_Cleaned.csv` file.
- Ensure dependencies:** Make sure you have `streamlit`, `pandas`, `plotly`, `matplotlib`, `seaborn`, `numpy` installed (`pip install -r requirements.txt` or `pip install streamlit pandas plotly matplotlib seaborn numpy`).
- Run the app:** Open your terminal or command prompt, navigate to the directory where you saved the file, and run:

```
streamlit run dashboard_app.py
```

4. **Access the dashboard:** Streamlit will open a new tab in your web browser with the dashboard.
5. **For cloud deployment:** Consider platforms like Streamlit Community Cloud or Hugging Face Spaces for wider stakeholder dissemination. You'll typically need to commit your `dashboard_app.py` and `Student_Performance_Analysis_Cleaned.csv` (and a `requirements.txt` file) to a GitHub repository.